

Q1. DIRECTIONS *for question 1:* Select the correct alternative from the given choices.

Instead of walking along the two adjacent sides of a rectangular tank, a boy swam across the tank along the diagonal. In doing so, he covered a distance which is lesser than the walking distance by a third of the longer side of the tank. Find the ratio of the length of the shorter side of the tank to that of the longer side.

☐ a) **8 : 15**

☐ b) **3 : 4**

☐ c) **7 : 24**

☐ d) **5 : 12**

Q2. DIRECTIONS *for question 2:* Type in your answer in the input box provided below the question.

If $36a^2 + \frac{1}{9a^2} = 4$, what is the value of $216a^3 - \frac{1}{27a^3}$.

Q3. DIRECTIONS *for question 3:* Select the correct alternative from the given choices.

What is the remainder when $28! + 27^{28}$ is divided by 29?

☐ a) 0

☐ b) 13

☐ c) 2

☐ d) 28

Q4. DIRECTIONS *for question 4:* Type in your answer in the input box provided below the question.

There are two geometric progressions, G_1 and G_2 , whose first terms are 4 and 8 respectively and whose common ratios are $2^{\frac{3}{2}}$ and $2^{\frac{7}{4}}$ respectively. If the last term of both progressions is 2^{289} , find the number of terms common to G_1 and G_2 .

Q5. DIRECTIONS *for questions 5 and 6:* Select the correct alternative from the given choices.

A function f is defined on integers in the following manner:

$$f(x) = 2x, \text{ if } x \text{ is odd.}$$

$$f(x) = x - 3, \text{ if } x \text{ is even.}$$

If $A_0 = 13$ and $A_{n+1} = 2[f(A_n)]$, find A_6 .

- ☐ a) 1484
- ☐ b) 1478
- ☐ c) 1490
- ☐ d) None of the above

Q6. DIRECTIONS *for questions 5 and 6:* Select the correct alternative from the given choices.

PQRS is a quadrilateral whose diagonals are perpendicular to each other. If PQ = 16 cm, QR = 12 cm and RS = 20 cm, find PS.

- ☐ a) $12\sqrt{2}$ cm.
- ☐ b) $8\sqrt{2}$ cm.
- ☐ c) $16\sqrt{2}$ cm.
- ☐ d) $24\sqrt{2}$ cm.

Q7. DIRECTIONS *for question 7:* Type in your answer in the input box provided below the question.

If the sum of the series $\frac{1}{12} + \frac{1}{20} + \frac{1}{30} + \frac{1}{42} + \dots \infty$ is equal to S , find the value of $72S$.

Q8. DIRECTIONS *for questions 8 to 11:* Select the correct alternative from the given choices.

Harshita went to the local market at a certain time when the angle between the minute hand and the hour hand of the wall clock in her room was 55° . She returned home and observed that the angle between the two hands of the clock was once again 55° . If she returned home in less than 45 minutes, for how long was she outside her home?

☐ a) $16\frac{3}{11}$ minutes

☐ b) **24 minutes**

☐ c) $22\frac{3}{11}$ minutes

☐ d) **20 minutes**

Q9. DIRECTIONS *for questions 8 to 11:* Select the correct alternative from the given choices.

How many integral values of x satisfy the equation $3^{\sqrt{x}} + 4^{\sqrt{x}} = 5^{\sqrt{x}}$?

☐ a) 0

☐ b) 1

☐ c) 2

☐ d) 3

Q10. DIRECTIONS *for questions 8 to 11:* Select the correct alternative from the given choices.

This morning, all the students of our school assembled for the assembly. The students stood in an odd number of rows with as many students per row as the number of rows. After the assembly, when they were all given chocolates. Each student was given exactly 10 chocolates. The chocolates given were distributed from packets of chocolates, each packet containing exactly 240 chocolates. After distribution was found that exactly one packet remained with some chocolates in it. Which of the following can be the number of chocolates remaining in this packet?

☐ a) 180

☐ b) 150

☐ c) 160

☐ d) 220

Q11. DIRECTIONS *for questions 8 to 11:* Select the correct alternative from the given choices.

How many points on the line $2x + 3y = 6$ are equidistant from the coordinate axes?

- ☐ a) 0
- ☐ b) 1
- ☐ c) 2
- ☐ d) More than 2

Q12. DIRECTIONS *for questions 12 to 15:* Type in your answer in the input box provided below the question.

The average weight of five persons in a group was 76 kg. After one year, the average weight of the same group became 81 kg. If each person in the group gained at least 2 kg, what is the maximum weight (in kg) that any person in the group could have gained?

Q13. DIRECTIONS *for questions 12 to 15:* Type in your answer in the input box provided below the question.

Six people have to go to attend an event in three different vehicles each of which can accommodate a maximum of six persons. In how many ways can they go, so that they use all the three vehicles?

Q14. DIRECTIONS *for questions 12 to 15:* Type in your answer in the input box provided below the question.

Find the area (in sq. units) enclosed by the curves $f(x) = |x - 1| - 1$ and $g(x) = 1 - |x - 1|$.

Q15. DIRECTIONS *for questions 12 to 15:* Type in your answer in the input box provided below the question.

If two sides of a triangle measure 6 cm and 10 cm, how many integral values are possible for the length (in cm) of the third side?

Q16. DIRECTIONS *for questions 16 to 21:* Select the correct alternative from the given choices.

If $2a = 3b = 5c$ and $4c = 5d = 6e$, then $\frac{bcd}{ae} =$

☐ a) **a.**

☐ b) **b.**

☐ c) **c.**

☒ d) **d.** ✓ **Your answer is correct**

Q17. DIRECTIONS *for questions 16 to 21:* Select the correct alternative from the given choices.

In a school, a chess tournament was conducted, in which, all the contestants were boys, except two girls. Each contestant played exactly once against each of the other contestants. The two girls together scored a total of 8 points. Among the boys, each boy scored the same number of points. In the tournament, a contestant gets one point for a win, half a point for a tie, and no points for a loss. Find the number of boys who participated in the tournament.

- ☐ a) 14 or 7
- ☐ b) 7 or 11
- ☐ c) 11 or 5
- ☐ d) **Cannot be determined**

Q18. DIRECTIONS *for questions 16 to 21:* Select the correct alternative from the given choices.

Find the difference between the maximum and the minimum possible values of a , for which the equation $3x^2 + ax - 3x + 12 = 0$ has real and equal roots.

☐ a) 9

☐ b) 15

☐ c) 6

☐ d) 24

Q19. DIRECTIONS *for questions 16 to 21:* Select the correct alternative from the given choices.

In an airport, a plane is scheduled to take off every 35 minutes, starting from 8:00 AM, and a plane is scheduled to land every 20 minutes, starting from 8:05 AM. Whenever a take-off and a landing are going to occur at the same time, the ground control will delay the plane that is about to land by three minutes, while all the subsequent planes will take-off/land on schedule. How many planes that are scheduled to land between 8:00 AM and 9:30 PM will get delayed?

☐ a) 5

☐ b) 3

☐ c) 2

☐ d) 6

Q20. DIRECTIONS for questions 16 to 21: Select the correct alternative from the given choices.

If $x = \frac{1}{4 - \left(\frac{1}{4 + \left(\frac{1}{4 - \left(\frac{1}{4 + \dots \infty} \right)} \right)} \right)}$, find the value of x .

- ☐ a) $\frac{\sqrt{65} - 7}{4}$
- ☐ b) $\frac{-7 - \sqrt{65}}{4}$
- ☐ c) $7 - 3\sqrt{5}$
- ☐ d) $3\sqrt{5} - 7$

Q21. DIRECTIONS *for questions 16 to 21:* Select the correct alternative from the given choices.

One can go from city A to city C either directly or via city B. If one can go from A to C in a total of 32 ways, from A to B in eight ways and from A to C directly in eight ways, then in how many ways can one go from B to C?

☐ a) 2

☐ b) 3

☐ c) 4

☐ d) 6

Q22. DIRECTIONS *for question 22:* Type in your answer in the input box provided below the question.

For any two sets A and B , $A \Delta B$ is defined as the set of elements which belong to either of the sets A and B but not both. X , Y and Z are subsets of the set of natural numbers, containing all the multiples of 2, 3 and 5 respectively, that are not greater than 100. Find the number of elements in $(X \Delta Y) \Delta Z$.